

Low-Rank Decay: Decoupled Spectral Decay for Grokking and Post-Pretraining Consolidation

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Preliminary Draft

Abstract

Modern Transformer architectures frequently employ normalization mechanisms such as RMSNorm and Query-Key Normalization, making parts of the model approximately scale-invariant with respect to weight magnitudes. In this regime, AdamW-style decoupled weight decay performs *radial decay*: it shrinks the Frobenius scale of a parameter matrix, but may not directly reduce the functional complexity of a scale-invariant layer. We propose *Low-Rank Decay* (LRD), an optimizer-side *decoupled spectral decay* step that can be applied on top of SGD, AdamW, or other base optimizers. Instead of decaying the matrix itself, LRD decays the polar factor $UV^\top$ of $W = U\Sigma V^\top$, producing a nuclear-norm-like shrinkage of singular values. We implement this step using a small number of Newton–Schulz iterations, avoiding explicit SVD in the training loop. On modular-addition grokking, preliminary experiments suggest that LRD induces effective-rank collapse in Query/Key matrices before delayed generalization, and may expand the low-data regime in which grokking occurs. We further argue that LRD should not be viewed as a universal pretraining optimizer. Rather, it is better understood as a stage-specific spectral-consolidation mechanism for mid-training, domain supervised fine-tuning, or chain-of-thought fine-tuning, where broad representations already exist and the goal is to suppress memorization-like high-rank directions while consolidating task-relevant circuits. This draft reports early evidence and outlines future experiments needed to establish robustness, causality, and connections to geometry-preserving model merging.

Keywords. Grokking; scale invariance; optimizer-side regularization; decoupled spectral decay; nuclear norm; Newton–Schulz iteration; rank collapse; training dynamics.

1 Introduction

Grokking refers to a delayed generalization phenomenon in which a model first memorizes the training set and only much later transitions to near-perfect test accuracy on algorithmic tasks [6]. This behavior suggests that training dynamics are not merely fitting labels, but may involve a transition from a high-capacity memorization solution to a lower-complexity algorithmic circuit.

At the same time, modern neural architectures increasingly rely on normalization layers. In Transformers, RMSNorm [7] and Query-Key Normalization (QK-Norm) normalize activations or attention queries/keys, making the resulting computation insensitive to scalar rescaling of certain weight matrices. This raises a basic question: *if a layer is scale-invariant, what does weight decay actually decay?*

AdamW-style weight decay can be viewed as *decoupled radial decay*: after a base optimizer step, it contracts the parameter matrix toward the origin. For a matrix W , this approximately shrinks all singular values proportionally, preserving the relative spectral shape. However, if the layer is invariant under $W \mapsto \alpha W$ for $\alpha > 0$, shrinking $\|W\|_F$ does not directly simplify the represented function. We therefore ask whether scale-invariant matrix parameters should be regularized by decaying their *singular spectrum* rather than their radial scale.

This draft introduces *Low-Rank Decay* (LRD), a decoupled spectral decay step that can wrap a standard optimizer. Given a base optimizer update, LRD applies an additional step

$$W_{t+1} = W_{t+1/2} - \eta_t \lambda \text{polar}(W_{t+1/2}), \quad (1)$$

where $W_{t+1/2}$ is the post-base-optimizer weight and $\text{polar}(W) = UV^\top$ for $W = U\Sigma V^\top$. This is the optimizer-side analogue of nuclear-norm shrinkage: it approximately subtracts a constant from each nonzero singular value, allowing small singular directions to disappear.

The intended positioning is important. We do *not* claim that LRD should replace pretraining optimizers. Early pretraining must preserve broad representation capacity and stable exploration; aggressive spectral collapse may remove directions that later become useful. Instead, we view LRD as a *post-pretraining spectral consolidation* mechanism: during mid-training, domain supervised fine-tuning (SFT), or chain-of-thought (CoT) SFT, the model already has broad representations, and the objective is often to consolidate task-relevant structure while reducing memorization-like high-rank traces.

Our preliminary experiments on modular addition suggest that LRD induces rapid effective-rank collapse in attention Query/Key matrices and improves grokking under low data fractions. The main contributions of this draft are:

1. We reformulate Low-Rank Decay as an optimizer-side decoupled spectral decay mechanism, analogous to AdamW’s decoupled radial decay but acting through nuclear-norm geometry.
2. We analyze singular-value dynamics, showing why radial decay preserves relative spectra while polar-factor decay performs soft-thresholding-like spectral shrinkage.
3. We present preliminary evidence that LRD induces rank collapse before generalization in modular arithmetic grokking.
4. We propose a stage-specific view: spectrum-preserving or update-balancing methods are better suited for early pretraining, LRD is better suited for mid-training/SFT consolidation, and geometry-preserving merging methods are natural tools after task capabilities have formed.

2 Background and Related Work

Grokking. Grokking was introduced by Power et al. [6] on algorithmic datasets such as modular arithmetic. Later work connected grokking to mechanistic interpretability, showing that generalization can correspond to the formation of structured circuits, for example Fourier or trigonometric circuits in modular addition [5]. Our work studies an optimizer-side intervention that may accelerate or stabilize the transition toward such lower-complexity solutions.

Scale-invariant optimization. A parameter block is scale-invariant if multiplying it by a positive scalar does not change the represented function. Prior work on normalization and scale-invariant parameters has shown that weight decay can alter effective learning rates by shrinking parameter

norms, even when it does not directly impose a functional complexity penalty [1, 4]. We use this as motivation to consider spectral decay instead of purely radial decay.

Decoupled weight decay. AdamW separates weight decay from the loss-gradient update and applies it as an optimizer-side step [?]. We adopt the same design philosophy, but replace decoupled radial decay with decoupled spectral decay. In this sense, LRD can be summarized as: *AdamW decays the parameter; LRD decays the polar factor.*

Spectral regularization and low-rank bias. The singular spectrum of a weight matrix is a natural complexity measure. Nuclear-norm penalties encourage low-rank solutions and are widely used as convex surrogates for rank. In deep learning, low-rank adaptation methods such as LoRA impose a hard architectural rank constraint [3]. In contrast, LRD is a soft optimizer-side bias: it encourages rank reduction but allows task gradients to maintain or regrow singular directions when needed.

Geometry-preserving adaptation and merging. Orthogonal fine-tuning and orthogonal model merging methods study transformations that preserve geometric structure in weight or representation space. OrthoMerge, for example, motivates model merging on orthogonal-group geometry and decomposes general updates into orthogonal and residual components [?]. This is complementary to LRD: OrthoMerge aims to preserve geometry after task-specific capabilities have formed, while LRD aims to reshape spectra during training or fine-tuning.

3 Scale Invariance and Radial Decay

Definition 1 (Scale invariance). *A function $f(x; W)$ is scale-invariant with respect to W if*

$$f(x; \alpha W) = f(x; W), \quad \forall \alpha > 0. \quad (2)$$

For an attention head with QK-Norm, attention logits take the schematic form

$$A(X) = \text{softmax} \left(\frac{\text{Norm}(XW_Q) \text{Norm}(XW_K)^\top}{\sqrt{d_h}} \right), \quad (3)$$

where $\text{Norm}(z)$ removes the scale of z . Hence scaling W_Q or W_K by a positive scalar does not change the normalized queries or keys.

Let $W = \rho\Theta$, where $\rho = \|W\|_F$ and $\|\Theta\|_F = 1$. Suppose the task loss $\mathcal{L}_{\text{task}}(W)$ depends only on Θ . Consider the continuous-time objective

$$J(W) = \mathcal{L}_{\text{task}}(W) + \frac{\lambda}{2} \|W\|_F^2. \quad (4)$$

The gradient flow is

$$\dot{W} = -\nabla_W \mathcal{L}_{\text{task}}(W) - \lambda W. \quad (5)$$

Since $\mathcal{L}_{\text{task}}$ is homogeneous of degree zero with respect to W , Euler’s homogeneous function theorem gives

$$\langle \nabla_W \mathcal{L}_{\text{task}}(W), W \rangle = 0. \quad (6)$$

Thus the task gradient is tangent to the sphere of constant Frobenius norm, while the weight-decay term is radial.

Proposition 1 (Radial decoupling of Frobenius weight decay). *Under the idealized assumptions above, Frobenius-norm weight decay affects only the radial scale ρ and has no direct projection onto the angular dynamics of Θ .*

Sketch. The radial dynamics satisfy

$$\dot{\rho} = \langle \dot{W}, \Theta \rangle = -\langle \nabla_W \mathcal{L}_{\text{task}}, \Theta \rangle - \lambda \langle W, \Theta \rangle = -\lambda \rho. \quad (7)$$

For the angular variable, the projection of $-\lambda W = -\lambda \rho \Theta$ onto the tangent space at Θ vanishes. Hence the decay term changes ρ but not the angular direction directly. $\square$

Remark 1. *This is an idealized continuous-time statement. In practical training with AdamW, finite steps, non-normalized parameters, and approximate scale invariance, weight decay can still influence optimization indirectly. The claim is not that weight decay is useless, but that radial decay is not a direct functional complexity prior for perfectly scale-invariant parameter blocks.*

4 Low-Rank Decay as Decoupled Spectral Decay

4.1 Base optimizer plus spectral decay

Let BaseOpt denote a standard optimizer such as SGD or AdamW. We write a base optimizer step as

$$W_{t+1/2} = \text{BaseOpt}(W_t, \nabla_W \mathcal{L}_t, \eta_t). \quad (8)$$

Low-Rank Decay then applies a separate spectral decay step:

$$W_{t+1} = W_{t+1/2} - \eta_t \lambda \widehat{\text{polar}}(W_{t+1/2}), \quad (9)$$

where $\widehat{\text{polar}}(W)$ is a Newton–Schulz approximation of the polar factor $UV^\top$.

This turns LRD into an optimizer wrapper rather than a loss-only penalty. The nuclear-norm objective

$$\mathcal{L}_{\text{task}}(W) + \lambda \|W\|_* \quad (10)$$

serves as a variational motivation, but the implemented algorithm is decoupled from the task-gradient step, mirroring the logic of AdamW.

4.2 Singular-value dynamics: radial versus spectral decay

Let $W = U\Sigma V^\top$, with singular values $\{\sigma_i\}$. Ignoring task gradients, decoupled radial decay updates

$$W^+ = W - \eta\lambda W = (1 - \eta\lambda)W. \quad (11)$$

Thus each singular value evolves as

$$\sigma_i^+ = (1 - \eta\lambda)\sigma_i. \quad (12)$$

Radial decay shrinks all singular values proportionally and largely preserves the relative spectral shape.

In contrast, idealized LRD updates

$$W^+ = W - \eta\lambda UV^\top = U(\Sigma - \eta\lambda I)V^\top, \quad (13)$$

Algorithm 1 LRD-BaseOpt: Decoupled Spectral Decay via Newton–Schulz

Require: Weight matrix W_t , stochastic gradient g_t , base optimizer BaseOpt, learning rate η_t , decay coefficient λ , iterations K , small $\epsilon > 0$

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1:  $W_{t+1/2} \leftarrow \text{BaseOpt}(W_t, g_t, \eta_t)$ 
2:  $X \leftarrow W_{t+1/2} / (\|W_{t+1/2}\|_F + \epsilon)$ 
3: for  $k = 1$  to  $K$  do
4:    $A \leftarrow X^\top X$ 
5:    $X \leftarrow \frac{1}{2}X(3I - A)$ 
6: end for
7:  $W_{t+1} \leftarrow W_{t+1/2} - \eta_t \lambda X$ 
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for nonzero singular values before thresholding. Hence

$$\sigma_i^+ \approx \max(\sigma_i - \eta\lambda, 0), \quad (14)$$

which resembles singular-value soft-thresholding. Small singular values are removed first, so LRD changes spectral shape and can induce rank selection.

This gives the central slogan:

AdamW performs decoupled radial decay; LRD performs decoupled spectral decay.

4.3 Newton–Schulz approximation

Exact SVD is expensive inside every optimizer step. We therefore approximate the polar factor using Newton–Schulz iteration [2]:

$$X_0 = \frac{W}{\|W\|_F + \epsilon}, \quad X_{k+1} = \frac{1}{2}X_k(3I - X_k^\top X_k). \quad (15)$$

For rectangular matrices, one can use the corresponding left- or right-oriented variant depending on matrix shape. In practice, a small fixed number of iterations is used. The method relies on matrix multiplications, which are efficient on modern accelerators.

4.4 Subdifferential geometry near low-rank strata

The nuclear norm becomes non-smooth at rank-deficient matrices. Let $W = U_r \Sigma_r V_r^\top$ be a compact SVD with rank r . The subdifferential is

$$\partial \|W\|_* = \left\{ U_r V_r^\top + Z : U_r^\top Z = 0, Z V_r = 0, \|Z\|_2 \leq 1 \right\}. \quad (16)$$

When W is square and full-rank, the orthogonal complements are trivial and the subgradient is unique. As rank decreases, the zero-singular-value subspaces enlarge, and the admissible Z directions expand. Geometrically, the subgradient changes from a near-unique “needle” direction to a fan-like set over the null singular subspaces.

This suggests the following interpretation of LRD. Early in training, when weights are effectively full-rank, LRD provides a clear polar-factor shrinkage direction. Later, near low-rank strata, the regularizer becomes less prescriptive: it has already removed many weak spectral directions, while task gradients can decide which remaining singular subspaces should be preserved or regrown. This gives LRD a soft-selection behavior rather than a hard fixed-rank constraint.

5 Stage-Specific View: Learning, Consolidation, and Composition

A central limitation of aggressive spectral decay is that it may remove useful directions too early. This motivates a stage-specific view of optimizer geometry.

Early pretraining: preserve capacity and stabilize exploration. During pretraining from scratch, the model must form broad representations over a heterogeneous data distribution. At this stage, one should avoid premature rank collapse. Geometry that preserves capacity, controls norms, balances updates, or maintains roughly isotropic exploration may be more appropriate than nuclear-norm-like shrinkage.

Mid-training and SFT: consolidate task structure. During continued training, domain SFT, or CoT SFT, the model already has broad representations. The risk shifts from under-exploration to memorization, spurious high-rank traces, or task-specific overfitting. Here LRD can act as a spectral bottleneck: it suppresses weak high-rank directions and encourages task-relevant knowledge or reasoning patterns to be consolidated into lower-dimensional structures.

Post-training merging: preserve learned geometry. After task-specific capabilities have formed, the goal of model merging is not to rediscover the task but to combine capabilities while avoiding interference. Geometry-preserving methods such as orthogonal/residual merging are natural at this stage. This suggests a high-level division of labor:

update-balancing methods for learning, LRD for consolidation, and geometry-preserving merging for composition.

This stage-specific framing is currently a hypothesis. The modular-addition experiments below serve as a small controlled setting for studying spectral consolidation, while future work should test domain SFT and CoT SFT settings directly.

6 Experimental Setup

Task. We use modular addition:

$$c = (a + b) \bmod p, \tag{17}$$

with $p = 113$ unless otherwise stated. The full dataset consists of all ordered pairs (a, b) .

Model. The model is a 2-layer Transformer with model dimension $d_{\text{model}} = 128$ and 4 attention heads. Pre-RMSNorm and QK-Norm are enabled to emphasize scale-invariant attention components.

Baselines. We compare standard Frobenius weight decay (L2/AdamW-style radial decay) against decoupled Low-Rank Decay. In the current preliminary experiments, LRD is primarily applied to Query and Key matrices, where QK-Norm makes the scale-invariance assumption most direct. [TODO: Specify exact parameter groups used in the final code.]

Placeholder for phase diagram: L2/AdamW radial decay vs. Low-Rank Decay across training data fraction and decay coefficient.

Figure 1: Preliminary phase diagram of modular-addition generalization. Replace this placeholder with the exported figure from the experiment notebook.

Placeholder for rank-collapse dynamics: stable rank and test accuracy over training steps.

Figure 2: Preliminary rank-collapse dynamics. Replace this placeholder with stable-rank traces for radial decay and LRD runs.

Metrics. We track train accuracy, test accuracy, loss, weight norms, gradient norms, singular values, and stable rank. For a matrix W , stable rank is defined as

$$\text{srank}(W) = \frac{\|W\|_F^2}{\|W\|_2^2}. \quad (18)$$

We report stable rank for Query/Key matrices during training.

Grid. We sweep training data fraction and decay strength to construct phase diagrams of generalization. A run is counted as successful grokking if it reaches test accuracy above 99%. [TODO: Add number of seeds and confidence intervals.]

Stage ablation. To test the stage-specific hypothesis, future experiments should compare LRD enabled from initialization, after warmup, at the onset of memorization, and only during a final consolidation window. If LRD is truly more suitable for consolidation than early learning, delayed activation should outperform always-on LRD in unstable regimes.

7 Preliminary Results

7.1 Phase diagram of generalization

Figure 1 is a placeholder for the phase diagram. Preliminary results suggest that standard L2 decay exhibits a generalization boundary around roughly half of the full data, while LRD expands the successful grokking region to lower data fractions. In the current draft, we treat these findings as preliminary because multi-seed variance, learning-rate matching, and stronger baselines are still needed.

7.2 Rank collapse dynamics

Figure 2 is a placeholder for effective-rank traces. In early experiments, LRD causes the stable rank of Query/Key matrices to decrease sharply before the rise in validation accuracy. This temporal ordering is consistent with the hypothesis that spectral compression precedes algorithm discovery. However, temporal precedence alone does not prove causality; causal interventions are needed.

8 Discussion

8.1 LRD as optimizer-side soft low-rank selection

Compared with hard low-rank parameterizations such as LoRA, LRD does not fix the rank in advance. Instead, it applies continuous pressure on the singular spectrum through an optimizer-side decay step. If a singular direction is useful for the task, the task gradient can counteract the nuclear-norm shrinkage. If a direction primarily supports memorization, it is more likely to be removed by the decay. This is the intended mechanism, but it remains to be tested with causal interventions.

8.2 Why not pretraining from scratch?

The stage-specific view is crucial. Early pretraining is a representation-formation process, and it may require high-dimensional capacity and stable exploration. Aggressive LRD from initialization could prematurely eliminate directions that later become useful. For this reason, we position LRD as a mid-training or SFT consolidation method rather than a universal pretraining optimizer.

This suggests practical schedules: no LRD during early training; mild LRD after a warmup; stronger LRD during a final consolidation phase; or LRD only on selected parameter groups such as attention Query/Key matrices under QK-Norm. These design choices should be evaluated empirically.

8.3 Connection to geometry-preserving model merging

A natural follow-up is to compare spectral-shaping during training with geometry preservation during model merging. Methods such as OrthoMerge preserve geometric structure after task-specific capabilities have formed. LRD instead reshapes the spectrum while the model is discovering or consolidating a circuit. This suggests a possible phase distinction: during training or SFT, it may be useful to collapse memorization-heavy spectral directions; after grokking, it may be useful to preserve the resulting low-dimensional algorithmic structure during merging.

One concrete future experiment is to train multiple small models that grok related modular tasks and then merge them using weight averaging versus an OrthoMerge-style orthogonal/residual decomposition. If grokked circuits are more geometrically composable than memorization solutions, LRD-trained models may merge more robustly.

8.4 Limitations

This draft has several important limitations:

1. Current evidence is based mainly on modular addition and should be extended to modular multiplication, permutation composition, and group operations.
2. Multi-seed results and error bars are needed to rule out instability or cherry-picking.
3. Learning-rate and effective-learning-rate controls are needed, since weight decay can indirectly change optimization speed in scale-invariant networks.
4. Exact SVD-based nuclear decay should be compared against Newton–Schulz LRD on small models.
5. Stage ablations are needed to test whether LRD is more effective as mid-training/SFT consolidation than as always-on pretraining decay.

6. Causal tests are needed to establish whether rank collapse causes grokking rather than merely correlating with it.

9 Conclusion

We presented Low-Rank Decay as a decoupled spectral decay mechanism for studying grokking and post-pretraining consolidation in scale-invariant Transformers. The central idea is that, when normalized layers remove sensitivity to radial weight scale, optimizer-side decay should not necessarily act on Frobenius magnitude. LRD instead decays the polar factor, producing nuclear-norm-like singular-value shrinkage and encouraging rank selection. Preliminary modular-addition experiments suggest that LRD induces rank collapse before delayed generalization and may expand the data-scarce grokking regime. The current evidence is promising but incomplete; the next step is to establish robustness, compare stage-specific schedules, perform causal interventions, and test whether LRD-consolidated circuits can be preserved and composed through geometry-aware model merging.

References

- [1] Sanjeev Arora, Nadav Cohen, and Elad Hazan. On the optimization of deep networks: Implicit acceleration by overparameterization. In *International Conference on Machine Learning*, 2018.
- [2] Nicholas J. Higham. *Functions of Matrices: Theory and Computation*. SIAM, 2008.
- [3] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2021.
- [4] Zhiyuan Li, Kaixuan Wei, and Sanjeev Arora. Exponential learning rates in scale-invariant neural networks. *arXiv preprint arXiv:2004.02584*, 2020.
- [5] Neel Nanda, Lawrence Chan, Tom Lieberum, Jess Smith, and Jacob Steinhardt. Progress measures for grokking via mechanistic interpretability. In *International Conference on Learning Representations*, 2023.
- [6] Alethea Power, Yuri Burda, Harri Edwards, Igor Babuschkin, and Vedant Misra. Grokking: Generalization beyond overfitting on small algorithmic datasets. *arXiv preprint arXiv:2201.02177*, 2022.
- [7] Biao Zhang and Rico Sennrich. Root mean square layer normalization. In *Advances in Neural Information Processing Systems*, 2019.